Group actions

Soham Saha

https://soham-saha.github.io

https://soham-saha.codeberg.page

- Given a group G and a set X, homomorphisms from G to $\mathrm{Sym}(X)$ are called group actions, and $\varphi \in \mathrm{Hom}(G, \mathrm{Sym}(X))$ is an action of G on X.
- Given $x \in X$, the stabilizer of x, denoted G_x or $\mathrm{Stab}(x)$ is defined as

$$G_x = \mathrm{Stab}(x) = \{g : (\varphi(g))(x) = x\}$$

  Intersection of all stabilizers is exactly the kernel of the group action,

$$\bigcap_{x \in X} G_x = \ker(\varphi) = \varphi^{-1}(\mathrm{id})$$

  All stabilizers are (not necessarily normal) subgroups of G, and therefore $\forall x, |\mathrm{Stab}(x)|$ divides $|G|$.
- Given $x \in X$, the orbit of x, denoted O_x is defined as

$$O_x = \{(\varphi(g))(x) : g \in G\}$$

  Orbits are not necessarily of same cardinality, they just partition X.
- For a general action $\varphi : G \to \mathrm{Sym}(X)$, the class equation is

$$X = \coprod_{\Omega \in \{O_x : x \in X\}} \Omega,$$

  which just tells that the orbits partition X. If we classify by whether orbit size is equal to 1 or not,

$$\begin{aligned} |X| &= |\{x : |O_x| = 1\}| + \sum_{\substack{\Omega \in \{O_x : x \in X\} \\ |\Omega| \neq 1}} |\Omega| \\ &= \text{number of fixed points} + \\ &\quad \text{total length of non-trivial orbits} \end{aligned}$$

  Class equation is all about partitioning the set into orbits.

- Think of the symmetry group of a tetrahedron acting on its set of vertices or, equivalently, S_4 acting on itself by left multiplication. This will give intuition for the bijection defined below. Also have a look at https://en.wikipedia.org/wiki/Group_action#Orbits_and_stabilizers.
- Given $x \in X$, we can define a set bijection (G/G_x is the set of left cosets) :

$$\begin{aligned} f : G/G_x &\to O_x \\ gG_x &\mapsto (\varphi(g))(x). \end{aligned}$$

  Each coset has a fixed place where it sends x to; f applied to a coset tells us what that place is.
- This gives us the orbit-stabilizer theorem,

$$|G_x||O_x| = |G|$$

Since $|G_x|$ divides $|G|$, so $|O_x|$ also divides $|G|$.

- If we have $\mathrm{Inn}(G) := \{(x \mapsto gxg^{-1}) : g \in G\}$, $\mathrm{Aut(G)}$ = isomorphisms on G, and $\mathrm{Sym}(G) = S_{\mathrm{ord}(G)}$ = group of all set theoretic permutations on G, then

$$\mathrm{Inn}(G) \triangleleft \mathrm{Aut}(G) \leq \mathrm{Sym}(G)$$

- Centralizers and conjugacy classes arise from the action of G on its own underlying set by conjugation. This action is defined as $\varphi_{\mathrm{conj}} : G \to \mathrm{Sym}(G) := g \mapsto (x \mapsto gxg^{-1})$.
- Given $x \in G$, the x-stabilizer of this action is called the centralizer of x, or $Z(x)$ for short. Therefore,

$$Z(x) = \{g : gxg^{-1} = x\}.$$

  Alternatively $Z(x)$ can be thought of as the set of all $g \in G$ which commute which x.
- Intersecting all the centralizers, we get the center of G, which is called $Z(G)$. These are the elements in G which commute with everyone in G. In the previous section we saw that the intersection of all stabilizers is the kernel of the action, which fits nicely here, since $\varphi_{\mathrm{conj}}^{-1}(\mathrm{id}) = Z(G)$. Since, $Z(G)$ is the kernel of φ, so $Z(G) \triangleleft G$.
- Repeating from previous point:

$$\ker(\varphi_{\mathrm{conj}}) = Z(G)$$

$$G/Z(G) \cong \mathrm{im}(\varphi_{\mathrm{conj}}) = \mathrm{Inn}(G) \leq \mathrm{Sym}(G)$$

- For all $x \notin Z(G)$ (non-central elements), $Z(x)$ properly contains $Z(G)$, since $Z(G) \subseteq Z(x)$ and $x \in Z(G) \setminus Z(x)$.
- Given $x \in G$, the orbit of x with respect to the conjugation action is called the conjugacy class of x. It is represented as $C(x)$ or $C_G(x)$. These are all the elements which can be reached by conjugating x with different things.

$$C(x) = \{gxg^{-1} : g \in G\}$$

  The order of $C(x)$ roughly tells about the "non-commutativity" of x, the higher the order, the less commutative x is.
- Elements of $Z(G)$ are precisely those which have singleton conjugacy classes.
- $|Z(x)|, |C(x)|, |Z(G)|$ all divide $|G|$. This is because $Z(x) \leq G$ and $Z(G) \triangleleft G$; and we also have $|Z(x)||C(x)| = |G|$ by the orbit-stabilizer theorem.
- The class equation for this specific action tells us that

$$G = Z(G) \sqcup \left(\coprod_{\substack{\Omega \in \{C(x) : x \in G\} \\ |\Omega| \neq 1}} \Omega \right)$$

$$\Rightarrow |G| = |Z(G)| + \sum_{\substack{\Omega \in \{C(x) : x \in G\} \\ |\Omega| \neq 1}} |\Omega|$$

  Remember that class equation is all about partitioning the set into orbits.

- Members of the same conjugacy class "behave similarly", meaning that they are equivalent elements in the group up to renaming the elements of the group. If x, y are in the same conjugacy

class, then $\exists g, gxg^{-1} = y$. g is the renaming operation on G; it renames the members of G while preserving all structure of the group.

- For $H, K \leq G$, if H, K are conjugate subgroups, then they "look alike" in G; they are the same subgroups in G up to renaming.
- The conjugation gxg^{-1} allows one to perform the "local action" x at a different location, dictated by g.

  For example, if $g = (1\ 2\ \ldots\ n)$, and $x = (1\ 2)$, then $g^n x g^{-n}$ allows us to do the "swap" x at a different place.
- Normal subgroups are union of conjugacy classes.

Key new ideas:

- Intersection of all stabilizers is the kernel of the action.
- Intersection of all centralizers is the center. Therefore, $Z(G) = \ker(\varphi_{\text{conj}})$.
- $G/Z(G) = G/\ker(\varphi_{\text{conj}}) \cong \text{im}(\varphi_{\text{conj}}) = \text{Inn}(G)$.
- For all $x \notin Z(G)$ (non-central element), $Z(x)$ properly contains $Z(G)$, i.e., $Z(x) \subsetneq Z(G)$.
- Normal subgroups are union of conjugacy classes.
- If there is a single subgroup of a given order, then it is normal; since if we have $H \leq G$ then $\forall g \in G, gHg^{-1} = H$.

Given $H \leq G$, some interesting group actions to consider are:

- G acting on itself by left multiplication.
- G acting on the set of left cosets of H by left multiplication.
- H acting on the set of left cosets of H by left multiplication.
- G acting on itself by conjugation.
- G acting on its sets of subsets by conjugation.
- G acting on the set of Sylow-p subgroups of G (also called $\text{Syl}_p(G)$) by conjugation.
- H, a Sylow-p subgroup of G acting on the set of Sylow-p subgroups of G by conjugation.

Suppose $\varphi : G \to \text{Sym}(\{H : H \leq G\})$ is the action of G acting on its own set of subsets by conjugation.

In this case, for $H \leq G$,

$$\text{orbit}(H) = \text{subgroups in } G \text{ who are conjugate to } H,$$
$$\text{stab}(H) = \text{normalizer of } H \text{ in } G = N_G(H).$$

Therefore, we have

$$N_G(H) = \{g : gHg^{-1} = H\} = \{g : gH = Hg\}$$

- Why is it called **normalizer**? Because $N_G(H)$ is the largest subgroup of G in which H is normal.
- We have

$$H \triangleleft N_G(H) \leq G.$$

  Therefore, $|H|$ divides $|N_G(H)|$, which further divides $|G|$.

- $|\text{orbit}(H)||\text{stab}(H)| = |G|$. Therefore, we have

  $$(\text{number of conjugates of H}) \times |N_G(H)| = |G|.$$

  This also tells us that

  $$\text{number of conjugates of } H = \frac{|G|}{|N_G(H)|} \leq \frac{|G|}{|H|} = [G : H]$$

- By Sylow's 3rd theorem, we know that all Sylow-p subgroups are conjugates to each other. Therefore,

  $$\begin{aligned} n_p &= \text{number of Sylow-}p\text{ subgroups} \\ &= \text{number of conjugates of some Sylow-}p\text{ subgroup} \\ &= \frac{|G|}{|N_G(H)|}, \text{where } H \text{ is a Sylow-}p \text{ subgroup.} \end{aligned}$$

- To find out $N_G(H)$, it is enough to consider generators of H.
- The following are equivalent:
  - $H \triangleleft G$
  - $|\text{orbit}(H)| = 1$
  - $\text{orbit}(H) = \{H\}$
  - $N_G(H) = G$.